Yicheng Gu

Research Intern at Spellbrush | **Master's Student at Aalto University**

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SUMMARY

Research Deep learning and digital signal processing for AI music and audio: neural vocoders and codecs, singing voice and speech synthesis, music representation learning, and digital audio effects. Experienced in building open-source systems and large-scale datasets.

Music DJ and music producer working in Japanese anime rhythm-game-centered electronic music and Vocaloid music. Sound designer working in films and games.

EDUCATION

Aalto University — Espoo, Finland
Master of Science in Speech and Language Technology (Major) — Sep. 2026 – Present
Master of Science in Acoustics and Audio Technology (Minor)
Master of Arts in Sound in New Media (Minor)
Supervisor: Prof. Lauri Juvela

Aalto University — Espoo, Finland
Exchange Student in Computer Science and Engineering — Sep. 2024 – Jun. 2025
Supervisor: Prof. Lauri Juvela

The Chinese University of Hong Kong — Shenzhen and Hong Kong
Bachelor of Engineering in Computer Science and Engineering — Sep. 2022 – Jun. 2026
Supervisor: Prof. Zhizheng Wu

EXPERIENCE

Spellbrush — San Francisco, CA, USA
Research Intern (Remote) · Project Leader: Tiger Tang — Jun. 2025 – Present

- **Neural audio synthesis:** Built Pupu-Vocoder and Pupu-Codec, the first aliasing-free neural vocoder and codec, outperforming existing systems on singing voice, music, and audio; code and checkpoints are open-source [J1].
- **Music representation learning:** Built PupuM2D, a Masked Modeling Duo (M2D) model trained on 2D music spectrograms that outperforms sequence-based self-supervised models on the MARBLEv2 benchmark [P3].
- **Product launch:** Contributed the text-to-speech (TTS) system to NORI_OS, a WebARG game for Nori that drew more than 10k players, over 1M trailer views, and 5k Steam wishlists. [Game] [Trailer] [Steam]

Acoustics Lab, Aalto University — Espoo, Finland
Visiting Scholar (On-Site) · Supervisor: Prof. Lauri Juvela — Sep. 2024 – Jun. 2025, Sep. 2026 – Present

- **Neural autotune:** Proposed Neurodyne, a state-of-the-art model benchmarked against Melodyne [C2].
- **Virtual analog modeling:** Built Solid State Bus-Comp, the first large-scale and diverse dataset for modeling the classic SSL 500 G-Bus compressor, used by many research institutions [C3].

Human Language Technology Lab, School of Data Science, CUHK — Shenzhen, China
Research Assistant (Hybrid) · Supervisor: Prof. Zhizheng Wu — Oct. 2022 – Jun. 2026

- **Open-source toolkit:** Co-founded Amphion, an open-source toolkit for audio, music, and speech generation used by researchers and developers worldwide, with 10k+ GitHub stars and 850+ forks [C4].
- **Neural vocoders:** Designed discriminators based on the constant-Q and continuous wavelet transforms (CQT, CWT) that improve GAN-based vocoders [C7, J4]; the CQT discriminator is implemented in NVIDIA's BigVGAN-v2.
- **Speech generation:** Co-developed Emilia, a large-scale, multilingual, and diverse in-the-wild speech generation dataset adopted by research institutions and companies worldwide, including NVIDIA [C5, J3].
- **Singing voice:** Investigated the complementary roles of content features for singing voice conversion [C8, C6].

Shanghai AI Laboratory — Shanghai, China
Research Assistant (Remote) · Supervisor: Dr. Yanhong Zeng — Dec. 2023 – Mar. 2025

- **Video-to-audio generation:** Integrated an IP-Adapter and a sound event detection model into an existing audio generation pipeline, achieving both audio-visual synchronization and text controllability in FoleyCrafter [J2].

Moises — Utah, USA
Research Collaborator (Remote) · Collaborator: Joaquim Cavalcante — Sep. 2025 – Present

- **Neural autotune:** Built an aliasing-free neural pitch-manipulation system and an autotune system that automatically corrects the pitch contour of a singing voice according to its musical context [P2].

OPEN-SOURCE PROJECTS

Amphion — 10k+ GitHub stars
Co-Founder · An open-source toolkit for audio, music, and speech generation

- Co-founded the toolkit [C4]; migrated and adapted widely used and state-of-the-art neural vocoders into it.
- Integrated a comprehensive suite of objective evaluation metrics into the framework.
- Reached #1 on GitHub Trending; its design was drawn on by the popular So-VITS-SVC 5.0 project.
- Covered by Synced, MarkTechPost, Shenzhen Special Zone Daily, and the Greater Bay Area News Agency; VC Jason Calacanis rated it "A+++ … This is the future!" on *This Week in Startups*.

YuE2 — 12k+ GitHub stars
Contributor · A frontier-quality open-source song generation model

- Contributed to the development of the VAE that decodes YuE2's acoustic latents into 48 kHz stereo audio [P1].
- YuE2 reached #1 on GitHub Trending (all languages), #3 on Hugging Face global trending, and #1 on its text-to-audio trending (Sep. 2026), and is competitive with Suno v5/v6 on WildSongBench.

FoleyCrafter — 600+ GitHub stars
Contributor · An OpenMMLab framework that brings silent videos to life with synchronized sounds

- Co-developed a video-to-audio model with audio-visual synchronization and text control [J2].

PUBLICATIONS

* Equal contribution · Google Scholar: 778 citations, h-index 8, i10-index 8 (as of October 2026)

Journal Articles

[J1] **Yicheng Gu**, Junan Zhang, Chaoren Wang, Jerry Li, Zhizheng Wu, Lauri Juvela. "Aliasing-Free Neural Audio Synthesis." *IEEE TASLP*, 2026. [Paper] [Code] [Demo]

[J2] Yiming Zhang, **Yicheng Gu**, Yanhong Zeng, Zhening Xing, Yuancheng Wang, …, Kai Chen. "FoleyCrafter: Bring Silent Videos to Life with Lifelike and Synchronized Sounds." *IJCV*, 2026. [Paper] [Code] [Demo]

[J3] Haorui He*, Zengqiang Shang*, …, **Yicheng Gu**, …, Zhizheng Wu. "Emilia: A Large-Scale, Extensive, Multilingual, and Diverse Dataset for Speech Generation." *IEEE TASLP*, 2025. [Paper] [Code] [Dataset]

[J4] **Yicheng Gu**, Xueyao Zhang, Liumeng Xue, Haizhou Li, Zhizheng Wu. "An Investigation of Time-Frequency Representation Discriminators for High-Fidelity Vocoders." *IEEE/ACM TASLP*, 2024. [Paper] [Code] [Demo]

Conference and Workshop Papers

[C1] Zirui Li, Jens Edlund, **Yicheng Gu**, Nhan Phan, Lauri Juvela, Mikko Kurimo. "Nord-Parl-TTS: Finnish and Swedish TTS Dataset from Parliament Speech." *ICASSP*, 2026. [Paper] [Demo] [Dataset]

[C2] **Yicheng Gu**, Chaoren Wang, Zhizheng Wu, Lauri Juvela. "Neurodyne: Neural Pitch Manipulation with Representation Learning and Cycle-Consistency GAN." *Interspeech*, 2025. [Paper] [Demo]

[C3] **Yicheng Gu***, Runsong Zhang*, Lauri Juvela, Zhizheng Wu. "Solid State Bus-Comp: A Large-Scale and Diverse Dataset for Dynamic Range Compressor Virtual Analog Modeling." *DAFx*, 2025. [Paper] [Demo] [Dataset]

[C4] Xueyao Zhang*, Liumeng Xue*, **Yicheng Gu***, Yuancheng Wang*, Jiaqi Li*, Haorui He, …, Zhizheng Wu. "Amphion: An Open-Source Audio, Music, and Speech Generation Toolkit." *SLT*, 2024. [Paper] [Code]

[C5] Haorui He*, Zengqiang Shang*, …, **Yicheng Gu**, …, Zhizheng Wu. "Emilia: An Extensive, Multilingual, and Diverse Speech Dataset for Large-Scale Speech Generation." *SLT*, 2024. [Paper] [Demo] [Dataset]

[C6] Xueyao Zhang, Zihao Fang, **Yicheng Gu**, Haopeng Chen, Lexiao Zou, ..., Zhizheng Wu. "Leveraging Diverse Semantic-based Audio Pretrained Models for Singing Voice Conversion." *SLT*, 2024. [Paper] [Demo]

[C7] **Yicheng Gu**, Xueyao Zhang, Liumeng Xue, Zhizheng Wu. "Multi-Scale Sub-Band Constant-Q Transform Discriminator for High-Fidelity Vocoder." *ICASSP*, 2024. [Paper] [Demo]

[C8] Xueyao Zhang, **Yicheng Gu**, ..., Zhizheng Wu. "Leveraging Content-based Features from Multiple Acoustic Models for Singing Voice Conversion." *ML4Audio Workshop at NeurIPS*, 2023. [Paper] [Code] [Demo]

Preprints

[P1] Ruibin Yuan, Jiahao Pan, Junyan Jiang, Zhiyue Wu, Ziya Zhou, ..., **Yicheng Gu**, ..., Yike Guo. "YuE2: Unifying Symbolic and Audio Music Generation at Frontier Quality." Submitted to *ICLR* 2027. [Paper] [Code] [Demo]

[P2] Joaquim Cavalcante, **Yicheng Gu**, ..., Thais Gaudencio. "MIDIBack: Harmony-Aware Singing Pitch Correction via Joint Vocal-Accompaniment Symbolic Modeling." Submitted to *ICASSP* 2027. [Paper] [Demo]

[P3] **Yicheng Gu**, Junan Zhang, Jerry Li, Zhizheng Wu, Lauri Juvela. "Frequency-Aware Self-Supervised Music Representation Learning." Submitted to *IEEE TASLP*. [Paper] [Code] [Demo]

[P4] **Yicheng Gu**, Chaoren Wang, Junan Zhang, Xueyao Zhang, Zihao Fang, Haorui He, Zhizheng Wu. "SingNet: Towards a Large-Scale, Diverse, and In-the-Wild Singing Voice Dataset." *arXiv*, 2025. [Paper]

[P5] **Yicheng Gu**, Pablo Pérez Zarazaga, Chaoren Wang, Zhizheng Wu, ..., Lauri Juvela. "HiFi-Glot: High-Fidelity Neural Formant Synthesis with Differentiable Resonant Filters." *arXiv*, 2024. [Paper] [Code] [Demo]

ACADEMIC SERVICE

Conference Reviewer

- ICASSP 2025, 2026, 2027
- Interspeech 2026
- ICLR 2025
- IJCNN 2025, 2026, 2027

Journal Reviewer

- IEEE TASLP
- IEEE SPL

HONORS AND AWARDS

- Aalto University Excellence Scholarship (Top 1 applicant in major) 2026
- Academic Performance Scholarship, Class A (Top 1%) 2025
- Nobel Class (Top 1) 2024
- Academic Performance Scholarship, Class A (Top 1%) 2024
- Academic Performance Scholarship, Class B (Top 3%) 2023
- "LanHuaYing" Scholarship (Top 10 admitted students in Zhejiang Province) 2022
- University Entrance Scholarship (Top 1% in Zhejiang Province) 2022